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Rx only

CEFTRIAXONE-ZHIJUN 1.0G

Ceftriaxone Sodium Powder for Injection

DESCRIPTION

CEFTRIAXONE-ZHIJUN in sterile form is supplied in vials as almost white or yellowish crystalline powder equivalent to 1.0 g of ceftriaxone (as ceftriaxone sodium). The product can be mixed with 1% Lidocaine HCl Injection, Sterile Water for Injection, 0.9% Sodium Chloride Injection, 5% Dextrose Injection and 10% Dextrose Injection. The color of reconstituted solutions ranges from light yellow to amber, depending on the length of storage, concentration and diluent used.

PHARMACODYNAMICS

Microbiology

The bactericidal activity of ceftriaxone results from inhibition of cell wall synthesis. Ceftriaxone exerts in vitro activity against a wide range of gram-negative and gram-positive microorganisms. It is highly stable to most β-lactamases, both penicillinases and cephalosporinases, of gram-positive and gram-negative bacteria. Ceftriaxone is usually active against the following microorganisms in vitro and in clinical infections (see Indications):

Gram-Positive Aerobes:

Staphylococcus aureus (methicillin-sensitive), *Staphylococci* coagulase-negative, *Streptococcus pyogenes* (β-hemolytic, group A), *Streptococcus agalactiae* (β-hemolytic, group B), β-hemolytic *Staphylococci* (non-group A or B), *Streptococcus viridians* and *Streptococcus pneumoniae*.

Note: Methicillin-resistant staphylococci are resistant to cephalosporins, including ceftriaxone. Most strains of Group D streptococci and enterococci, eg, *Enterococcus (Streptococcus) faecalis* are resistant.

Gram-Negative Aerobes:

Acinetobacter lwoffii, *Acinetobacter anitratus* (mostly, *A. baumannii*)*, *Aeromonas hydrophila*, *Alcaligenes faecalis*, *Alcaligenes odorans*, *Alcaligenes*-like bacteria, *Borrelia burgdorferi*, *Capnocytophaga sp*, *Citrobacter diversus* (including *C. amalonaticus*), *Citrobacter freundii**, *Escherichia coli*, *Enterobacter aerogenes**, *Enterobacter cloacae**, *Enterobacter sp* (other)*, *Haemophilus ducreyi*, *Haemophilus influenzae*, *Haemophilus parainfluenzae*, *Hafnia alvei*, *Klebsiella oxytoca*, *Klebsiella pneumoniae****, *Moraxella catarrhalis* (formerly *Branhamella catarrhalis*), *Moraxella osloensis*, *Moraxella sp* (other), *Morganella morganii*, *Neisseria gonorrhoeae*, *Neisseria meningitidis*, *Pasteurella multocida*, *Plesiomonas shigelloides*, *Proteus mirabilis*, *Proteus penneri**, *Proteus vulgaris**, *Pseudomonas fluorescens**, *Pseudomonas sp* (other)*, *Providentia rettgeri**, *Providencia sp* (other), *Salmonella typhi*, *Salmonella sp* (non-typhoid), *Serratia marcescens**, *Serratia sp* (other)*, *Shigella* and *Vibrio spp*, *Yersinia enterocolitica* *Yersinia sp* (other).

* Some isolates of these species are resistant to ceftriaxone, mainly due to the production of the chromosomally encoded β-lactamase.

** Some isolates of these species are resistant due to production of extended spectrum, plasmid-mediated β-lactamase.

Note: Many strains of the above microorganisms that are multiple resistant to other antibiotics, eg, penicillins, cephalosporins, and aminoglycosides, are susceptible to ceftriaxone.

Anaerobic Organisms:

Bacteroides sp (bile-sensitive)*, *Clostridium sp* (excluding *C. difficile*), *Fusobacterium nucleatum*, *Fusobacterium sp* (other), *Gaffkia anaerobica* (formerly *Peptococcus*), *Peptostreptococcus sp*.

*Some isolates of these species are resistant to ceftriaxone due to β-lactamase-production.

Note: Many strains of β-lactamase-producing *Bacteroides sp* (notably *B. fragilis*) are resistant. *Clostridium difficile* is resistant.

PHARMACOKINETICS

The pharmacokinetics of ceftriaxone are nonlinear and all basic pharmacokinetic parameters, except the elimination half-life, are dose-dependent if based on total drug concentrations.

Absorption: The maximum plasma concentration after a single IM dose of 1 g is about 81 mg/L and is reached in 2-3 hrs after administration. The area under the plasma concentration-time curve after IM administration is equivalent to that after IV administration of an equivalent dose, indicating 100% bioavailability of IM-administered ceftriaxone.

Distribution: The volume of distribution of ceftriaxone is 7-12 L. Ceftriaxone has shown excellent tissue and body fluid penetration after a dose of 1-2 g; concentrations well above the minimal inhibitory concentrations (MICs) of most pathogens responsible for infection are detectable for >24 hrs in over 60 tissues or body fluids including lung, heart, biliary tract/liver, tonsil, middle ear and nasal mucosa, bone as well as cerebrospinal, pleural, prostatic and synovial fluids.

On IV administration, ceftriaxone diffuses rapidly into the intestinal fluid where bactericidal concentrations against susceptible organisms are maintained for 24 hrs. Protein-Binding: Ceftriaxone is reversibly bound to albumin and the binding decreases with the increase in the concentration eg, from 95% binding at plasma concentrations of <100 mg/L to 85% binding at 300 mg/L. Owing to the lower albumin content, the proportion of free ceftriaxone in interstitial fluid is correspondingly higher than in plasma.

Penetration Into Particular Tissues: Ceftriaxone penetrates the inflamed meninges of neonates, infants and children: Ceftriaxone concentrations exceed 1.4 mg/L in the cerebrospinal fluid (CSF) 24 hrs after IV injection of doses of 50-100 mg/kg (neonates and infants, respectively). Peak concentration in CSF is reached about 4 hrs after IV injection and gives an average value of 18 mg/L. Mean CSF levels are 17% of plasma concentrations in patients with bacterial meningitis.

In adult meningitis patients, administration of 50 mg/kg leads within 2-24 hrs to CSF concentrations several times higher than the MICs required for the most common meningitis pathogens.

Ceftriaxone crosses the placental barrier and is excreted in breast milk at low concentrations.

Metabolism: Ceftriaxone is not metabolized systematically; but is converted to inactive metabolites by the gut flora.

Elimination: Total plasma clearance is 10-22 mL/min. Renal clearance is 5-12 mL/min. 50-60% of ceftriaxone is excreted unchanged in the urine, while 40-50% is excreted unchanged in the bile. The elimination half-life in adults is about 8 hrs.

Pharmacokinetics in Special Clinical Situations: In neonates, urinary recovery accounts for about 70% of the dose. In infants aged <8 days and in elderly persons >75 years, the average elimination half-life is usually 2-3 times that in young adults.

In patients with renal or hepatic dysfunction, the pharmacokinetics of ceftriaxone are only minimally altered and the elimination half-life is only slightly increased. If kidney function alone is impaired, biliary elimination of ceftriaxone is increased; if liver function alone is impaired, renal elimination is increased.

INDICATIONS

CEFTRIAXONE-ZHIJUN is indicated in the treatment of the following infections caused by pathogens sensitive to ceftriaxone, e.g.:

- Sepsis;
- Meningitis;
- Infections of the bones, joints, soft tissue, skin and of wounds;
- Infections in patients with impaired defense mechanisms;
- Respiratory tract infections, particularly pneumonia;
- Genital infections, including gonorrhea;
- Peri-operative prophylaxis of infections associated with surgery.

DOSAGE & ADMINISTRATION

Adults and Children >12 years: Usual Dosage: 1-2 g once daily (every 24 hrs). In severe cases, the dosage may be raised to a maximum dose of 4 g, administered once daily.

Neonates, Infants and Children up to 12 years: The following dosage schedules are recommended for once-daily administration: **Neonates (up to 2 weeks):** A daily dose of 20-50 mg/kg body weight, not to exceed 50 mg/kg. **Infants and children (3 weeks to 12 years):** A daily dose of 20-80 mg/kg.

Elderly: The dosages recommended for adults require no modification in the case of geriatric patients.

Special Dosage Instructions:

Meningitis: In bacterial meningitis in infants and children, treatment begins with doses of 100 mg/kg (up to a maximum of 4 g) once daily. As soon as the causative organism has been identified and its sensitivity determined, the dosage can be reduced accordingly. The following duration of therapy has shown to be effective:

Neisseria meningitidis: 4 days; **Haemophilus influenzae:** 6 days and **Streptococcus pneumoniae:** 7 days.

Gonorrhea: A single IM dose of 250 mg ceftriaxone is recommended.

Impaired Renal and Hepatic Function: In patients with impaired renal function, there is no need to reduce the dosage of ceftriaxone provided hepatic function is intact. Only in cases of preterminal renal failure (creatinine clearance <10 mL/min) should the ceftriaxone dosage not exceed 2 g daily. In patients with liver damage, there is no need for the dosage to be reduced provided renal function is intact.

In patients with both severe renal and hepatic dysfunction, the plasma concentrations of ceftriaxone should be determined at regular intervals and, if necessary, the dose should be adjusted.

In patients undergoing dialysis, no additional supplementary dosing is required following dialysis. Plasma concentrations should, however, be monitored, to determine whether dosage adjustments are necessary, since the elimination rate in these patients may be altered.

Directions for use:

IM Injection: Ceftriaxone 1 g is dissolved in 3.6 mL of 1% Lidocaine HCl Injection and injected well within the body of a relatively large muscle. It is recommended that not more than 1 g be injected at 1 site. The Lidocaine solution must never be administered IV.

IV Injection: Ceftriaxone 1 g is dissolved in 10 mL of Sterile Water for Injection. The IV administration should be given over 2-4 min.

IV Infusion: The infusion should be given over at least 30 min. Ceftriaxone 2 g is dissolved in 40 mL of one of the following calcium-free infusion solutions: 0.9% Sodium Chloride Injection, 5% Dextrose Injection, 10% Dextrose Injection and Sterile Water for Injection. Ceftriaxone solutions should not be mixed with or piggybacked into solutions containing other antimicrobial drugs or into diluent solutions other than those listed previously, owing to possible incompatibility.

Do not use diluents containing Calcium, such as Ringer's Solution or Hartmann's Solution, to reconstitute Ceftriaxone. Particulate formation can result.

OVERDOSAGE

In case of overdosage, drug concentration would not be reduced by hemodialysis or peritoneal dialysis. There is no specific antidote. Treatment of overdosage should be symptomatic.

CONTRAINDICATIONS

Patients with known hypersensitivity to the cephalosporin class of antibiotics. In patients hypersensitive to penicillin, the possibility of allergic cross-reactions should be borne in mind.

Ceftriaxone is contraindicated in neonates (≤28 days of age) if they require (or are expected to require) treatment with calcium-containing intravenous solutions, including calcium-containing infusions such as parenteral nutrition, because of the risk of precipitation of ceftriaxone-calcium.

WARNINGS

Ceftriaxone must not be mixed or administered simultaneously with calcium containing solutions or products, even via different infusion lines. Calcium containing solutions or products must not be administered within 48 hours of last administration of ceftriaxone. Cases of fatal reactions with calcium ceftriaxone precipitates in lung and kidneys in both term and premature neonates have been described. In some cases the infusion lines and times of administration and calcium-containing solutions differed.

Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients receiving therapy with beta-lactams. Before initiating therapy with CEFTRIAXONE-ZHIJUN, careful inquiry should be made concerning previous hypersensitivity reactions to penicillins, cephalosporins, carbapenems or other beta-lactam agents. If an allergic reaction occurs, CEFTRIAXONE-ZHIJUN must be discontinued immediately and appropriate alternative therapy instituted.

- In patients other than neonates, Ceftriaxone and calcium-containing solutions may be administered sequentially to one another if the infusion lines are thoroughly flushed between infusions with a compatible fluid.

- Diluents containing calcium, such as Ringer's solution or Hartmann's solution, are not to be used to reconstitute Ceftriaxone vials or to further dilute a reconstituted vial for intravenous administration because a precipitate can form. Ceftriaxone must not be administered simultaneously with calcium-containing intravenous solutions, including continuous calcium-containing infusions such as parenteral nutrition via a Y-site, because precipitation of ceftriaxone-calcium can occur.

PRECAUTIONS

As with other cephalosporins, anaphylactic shock cannot be ruled out even if a thorough patient history is taken.

Pseudomembranous colitis has been reported with nearly all antibacterial agents, including ceftriaxone. Therefore, it is important to consider this diagnosis in patients who present with diarrhea subsequent to the administration of antibacterial agents.

Superinfections with nonsusceptible microorganisms may occur as with other antibacterial agents.

Shadows which have been mistaken for gallstones have been detected on sonograms of the gallbladder, usually following doses higher than the standard recommended dose. These shadows are, however, precipitates of calcium ceftriaxone which disappear on completion or discontinuation of ceftriaxone therapy. Rarely have these findings been associated with symptoms. In symptomatic cases, conservative nonsurgical management is recommended. Discontinuation of ceftriaxone treatment in symptomatic cases should be at the discretion of the physician.

Use in pregnancy: Although there is no experimental evidence of mutagenic or teratogenic effect, ceftriaxone should not be used in pregnancy

(particularly in the first trimester) unless absolutely indicated.

Use in lactation: Low concentrations of ceftriaxone are excreted in human milk. Caution should be exercised when ceftriaxone is administered to a nursing woman.

Use in children: Safety and effectiveness of ceftriaxone in neonates, infants and children have been established for the dosages described under Dosage & Administration. Studies have shown that ceftriaxone, like some other cephalosporins, can displace bilirubin from serum albumin. Therefore, caution should be exercised when considering ceftriaxone treatment in hyperbilirubinemic neonates. Ceftriaxone should not be used in neonates (especially premature) at risk of developing bilirubin encephalopathy.

ADVERSE REACTIONS

During the use of ceftriaxone, the following side effects, which were reversible either spontaneously or after withdrawal of the drug, have been observed: Gastrointestinal: Loose stools or diarrhea, nausea, vomiting, stomatitis and glossitis.

Hematological: Eosinophilia, leukopenia, granulocytopenia, hemolytic anemia, thrombocytopenia.

Skin reactions: Exanthema, allergic dermatitis, pruritus, urticaria and edema. Isolated cases of severe cutaneous adverse reactions (erythema multiforme, Stevens-Johnson syndrome or Lyell's syndrome/toxic epidermal necrolysis) have been reported.

Others: Rare: Headache and dizziness, symptomatic precipitation of ceftriaxone calcium salt in the gallbladder, increase in liver enzymes, oliguria, increase in serum creatinine, genital mycosis, shivering and anaphylactic or anaphylactoid reactions.

Pseudomembranous enterocolitis and coagulation disorders have been reported as very rare side effects.

Local: In rare cases, phlebitic reactions occurred after IV administration. These may be minimized by slow (2-4 min) injection.

Nervous system disorders: Frequency 'not known': Encephalopathy*

*Reversible encephalopathy has been reported with the use of ceftriaxone, particularly when high doses are administered in patients with renal impairment and additional predisposing factors such as older age, pre-existing central nervous system disorders.

INTERACTIONS

No impairment of renal function has so far been observed after concurrent administration of large doses of ceftriaxone and potent diuretics (eg, furosemide). There is no evidence that ceftriaxone increases the renal toxicity of aminoglycosides. No effect similar to that of disulfiram has been demonstrated after ingestion of alcohol subsequent to the administration of ceftriaxone.

The elimination of ceftriaxone is not altered by probenecid.

In an in vitro study, antagonistic effects have been observed with the combination of chloramphenicol and ceftriaxone.

Influence on Diagnostic Tests: In patients treated with ceftriaxone, Coombs' test may rarely become false-positive. Ceftriaxone, like other antibiotics, may result in false-positive tests for galactosemia.

Likewise, non-enzymatic methods for the glucose determination in urine may give false-positive results. For this reason, urine-glucose determination during therapy with ceftriaxone should be done enzymatically.

Incompatibilities

Ceftriaxone should not be added to solutions containing calcium eg, Hartmann's solution and Ringer's solution.

Based on literature reports, ceftriaxone is incompatible with ampicillin, vancomycin, fluconazole and aminoglycosides.

PACKAGING

10 vials/box.

STORAGE CONDITION

Dry powder: Store below 30°C and protect from light.

Reconstituted solutions: Reconstituted solutions retain their physical and chemical stability for 6 hrs at room temperature (below 25°C) or 24 hrs in the refrigerator (2-8°C).

 **Manufactured by:**
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名称	注射用头孢曲松钠说明书—马来西亚		
规格	通用	版本号	07
尺寸	143mm×232mm	设计	武广利
注册审核/日期	年 月 日		